

Douglas B. Markant

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Updated: May 18, 2026

EDUCATION

2008—2014	New York University Psychology: Cognition and Perception M.A. completed May 16, 2012 Ph.D. completed January 27, 2014
2001—2005	Cornell University B.A. Psychology Minors in Information Science and Cognitive Science

POSITIONS

2017—present	Associate Professor Department of Psychological Science University of North Carolina at Charlotte
2014—2017	Postdoctoral Researcher Center for Adaptive Rationality Max Planck Institute for Human Development, Berlin Advisors: Dr. Ralph Hertwig and Dr. Tim Pleskac
2005—2008	Research Assistant Psychiatry Neuroimaging Laboratory Brigham and Women's Hospital, Harvard Medical School, Boston Advisor: Dr. Martha Shenton

PEER-REVIEWED JOURNAL ARTICLES

*Underlined names denote student co-authors. * indicates shared first authorship.*

Markant, D. (2026). Asymmetric shape variability biases proportion judgments with icon arrays.

Cognitive Science 50(5), 1–29. doi: 10.1111/cogs.70227

[\[link to publisher\]](#)[\[code and data repository\]](#)

Markant, D., Battisti, A., and Ruggeri, A. (2026). Developmental changes in active learning of transitive relations. *Developmental Psychology*. doi: 10.1037/dev0002187

[\[link to publisher\]](#)[\[code and data repository\]](#)

Glass, S., Saxena, N., and **Markant, D.** (2026). Assessing Needs for Game-Based Cognitive Flexibility Training: Interview Study Among Therapists and Clients. *Journal of Cognitive Intervention and Digital Health*. doi: 10.58558/jcd.2025.4.2.29

[\[link to publisher\]](#)

Markant, D., Sah, S., Karduni, A., Rogha, M., Thai, M., and Dou, W. (2025). Correcting misperceptions at a glance: Using data visualizations to reduce political sectarianism. *IEEE Transactions on Visualization and Computer Graphics*. doi: 10.1109/TVCG.2025.3634777

[\[link to publisher\]](#)[\[preprint\]](#)[\[code and data repository\]](#)

Ruggeri, A., Li, Y-L., Battisti, A., and **Markant, D.** (2025). Active control of learning enhances memory across the lifespan. *Journal of Educational Psychology*.

[\[link to publisher\]](#)

Rogha, M., Sah, S., Karduni, A., **Markant, D.**, and Dou, W. (2024). The impact of elicitation and contrast charts on engagement, recall, and attitude change with news articles containing data visualizations. *IEEE Transactions on Visualization and Computer Graphics*. doi: 10.1109/TVCG.2024.3355884

[\[link to publisher\]](#)[\[preprint\]](#)[\[code and data repository\]](#)

***Markant, D.** and ***Galati, A.** (2023). Development of a Concept Inventory on Open and Transparent Research Practices. *Collabra: Psychology* 9(1), 75226. doi: 10.1525/collabra.75226

[\[link to publisher\]](#)[\[preprint\]](#) [\[code and data repository\]](#)

Markant, D. (2021). Chained study and the discovery of relational structure. *Memory and Cognition* 50, 95–111. doi: 10.3758/s13421-021-01201-1

[\[link to publisher\]](#) [\[data repository\]](#)

Wesslen, R., Karduni, A., **Markant, D.**, and Dou, W. (2021). Effects of uncertainty visualizations on myopic loss aversion and equity premium puzzle in retirement investment decisions. *IEEE Transactions on Visualization and Computer Graphics* 28(1), 454–464. doi: 10.1109/TVCG.2021.3114692

[\[link to publisher\]](#) [\[preprint\]](#) [\[code and data repository\]](#)

Jones, A., **Markant, D.**, Pachur, T., Gopnik, A., and Ruggeri, A. (2021). How is the hypothesis space represented? Evidence from young children’s active search and predictions in a multiple-cue inference task. *Developmental Psychology* 57(7), 1080–1093. doi: 10.1037/dev0001201

[\[link to publisher\]](#) [\[preprint\]](#)

Karduni, A., Markant, D., Wesslen, R., & Dou, W. (2021). A Bayesian cognition approach for belief updating of correlation judgment through uncertainty visualizations. *IEEE Transactions on Visualization and Computer Graphics* 27(2). doi: 10.1109/TVCG.2020.3029412

[\[link to publisher\]](#) [\[preprint\]](#)

Fantasia, V., **Markant, D.**, Valeri, G., Perry, N., & Ruggeri, A. (2020). Memory enhancements from active control of learning in children with Autism Spectrum Disorder. *Autism* 24(8), 1995–2007. doi: 10.1177/1362361320931244

[\[link to publisher\]](#)

Markant, D. (2020). Active transitive inference: When learner control facilitates integrative encoding. *Cognition* 200. doi: 10.1016/j.cognition.2020.104188

Ruggeri, A., **Markant, D.**, Gureckis, T. M., & Xu, F. (2019). Memory enhancements from active control of learning emerge across development. *Cognition* 186, 82–94. doi: 10.1016/j.cognition.2019.01.010

Markant, D. (2018). Effects of biased hypothesis generation on self-directed category learning. *Journal of Experimental Psychology: Learning, Memory, & Cognition*. doi: 10.1037/xlm0000671

Markant, D., Ruggeri, A., Gureckis, T. M., and Xu, F. (2016). Enhanced memory as a common effect of active learning. *Mind, Brain, and Education* 10(3):142–152. doi: 10.1111/mbe.12117

Markant, D., Settles, B. and Gureckis, T.M (2016). Self-directed learning favors local, rather than global, uncertainty. *Cognitive Science* 40(1), 100–120. doi: 10.1111/cogs.12220

Gureckis, T. M., Martin, J., McDonnell, J., Rich, A., **Markant, D.**, Coenen, A., Hamrick, J., Chan, P. (2015). psiTurk: An open-source framework for conducting replicable behavioral experiments online. *Behavior Research Methods*. doi: 10.3758/s13428-015-0642-8

Markant, D., Dubrow, S., Davachi, L., Gureckis, T.M. (2014). Deconstructing the effect of self-directed study on episodic memory. *Memory & Cognition* 42(8), 1211–1224. doi: 10.3758/s13421-014-0435-9

** Awarded the 2014 Psychonomic Society's Clifford T. Morgan Best Article Award for Memory & Cognition*

Markant, D. and Gureckis, T.M. (2014). Is it better to select or receive? Learning via active and passive hypothesis testing. *Journal of Experimental Psychology: General* 143(1), 94–122. doi: 10.1037/a0032108

Gureckis, T.M. and **Markant, D.** (2012). Self-directed learning: A cognitive and computational perspective. *Perspectives on Psychological Science*, 7(5), 464–481. doi: 10.1177/1745691612454304

Whitford, T. J., Kubicki, M., Schneiderman, J. S., O'Donnell, L. J., King, R., Alvarado, J. L., Khan, U., **Markant, D.**, Nestor, P. G., Niznikiewicz, M., et al. (2010). Corpus callosum abnormalities and their association with psychotic symptoms in patients with schizophrenia. *Biological Psychiatry*, 68(1):70–77.

Voineskos, A. N., O'Donnell, L. J., Lobaugh, N. J., **Markant, D.**, Ameis, S. H., Niethammer, M., Mulsant, B. H., Pollock, B. G., Kennedy, J. L., Westin, C. F., et al. (2009). Quantitative examination of a novel clustering method using magnetic resonance diffusion tensor tractography. *Neuroimage*, 45(2):370–376.

Rosenberger, G., Kubicki, M., Nestor, P. G., Connor, E., Bushell, G. B., **Markant, D.**, Niznikiewicz, M., Westin, C.-F., Kikinis, R., Saykin, A. J., et al. (2008). Age-related deficits in fronto-temporal connections in schizophrenia: a diffusion tensor imaging study. *Schizophrenia Research*, 102(1):181–188.

Kubicki, M., Styner, M., Bouix, S., Gerig, G., **Markant, D.**, Smith, K., Kikinis, R., McCarley, R., and Shenton, M. (2008). Reduced interhemispheric connectivity in schizophrenia-tractography based segmentation of the corpus callosum. *Schizophrenia Research*, 106(2):125–131.

PEER-REVIEWED CONFERENCE PROCEEDINGS

Markant, D. (2025). Self-verification and the perceived reliability of uncertain feedback sources. *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. San Francisco, CA: Cognitive Science Society.

Markant, D., Rogha, M., Karduni, A., Wesslen, R., and Dou, W. (2023). When do data visualizations persuade? The impact of prior attitudes on learning from visualizations of correlations. *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI 2023)*. doi: 10.1145/3544548.3581330
[\[preprint\]](#) [\[code and data repository\]](#)

Karduni, A., Wesslen, R., **Markant, D.**, and Dou, W. (2023). Images, Emotions, and Credibility: Effect of Emotional Facial Images on Perceptions of News Content Bias and Source Credibility in Social Media. *International AAAI Conference on Web and Social Media (ICWSM 2023)*. doi: 10.1609/icwsml.v17i1.22161
[\[preprint\]](#)

Markant, D., Rogha, M., Karduni, A., Wesslen, R., and Dou, W. (2022). Can data visualizations change minds? Identifying mechanisms of elaborative thinking and persuasion. *2022 IEEE Workshop on Visualization for Social Good (VIS4Good)*. IEEE, 2022.

Markant, D. (2022). Modeling the effect of chained study in transitive inference. *Proceedings of the 44th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society.
[\[link to paper\]](#)

Wesslen, R., **Markant, D.**, Karduni, A., & Dou, W. (2020). Using resource-rational analysis to understand cognitive biases in interactive data visualizations. VisPsych: IEEE VIS2020 Workshop Proceedings.

Markant, D. (2020). Effects of “chained” study on spontaneous relational discovery. In Denison, S., Mack, M., Xu, Yang., and Armstrong, B., editors, *Proceedings of the 42nd Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society.

Markant, D. (2019). Navigating the ‘chain of command’: Enhanced integrative encoding through active control of study. In Goel, V., Seifert, C., and Freska, C., editors, *Proceedings of the 41st Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society.

Perri, N., Fantasia, V., **Markant, D.**, De Simone, C., Valeri, G., & Ruggeri, A. (2019). Benefits of active control of study in autistic children. In Goel, V., Seifert, C., and Freska, C., editors, *Proceedings of the 41st Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society.

Adams, K., Kachergis, G., and **Markant, D.** (2017). Children's familiarity preference in self-directed study improves recognition memory. In G. Gunzelmann, A. Howes, T. Tenbrink, & E. J. Davelaar (Eds.), *Proceedings of the 39th Annual Conference of the Cognitive Science Society* (pp. 1501–1506). Austin, TX: Cognitive Science Society.

Markant, D. (2016). The impact of biased hypothesis generation on self-directed learning. In Papafragou, A., Grodner, D., Mirman, D., and Trueswell, J. (Eds.) *Proceedings of the 38th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society.

Ruggeri, A., **Markant, D.**, Gureckis, T. M., and Xu, F. (2016). Active control of study leads to improved recognition memory in children. In Papafragou, A., Grodner, D., Mirman, D., and Trueswell, J. (Eds.) *Proceedings of the 38th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society.

Markant, D., Pleskac, T. J., Diederich, A., Pachur, T., and Hertwig, R. (2015). Modeling choice and search in decisions from experience: A sequential sampling approach. In Noelle, D. C., Dale, R., Warlaumont, A. S., Yoshimi, J., Matlock, T., Jennings, C. D., & Maglio, P. P. (Eds.), *Proceedings of the 37th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society.

Markant, D. and Gureckis, T.M. (2014). A preference for the unpredictable over the informative during self-directed learning. In P. Bello, M. Guarini, M. McShane, & B. Scassellati (Eds.) *Proceedings of the 36th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society.

Markant, D. and Gureckis, T.M. (2012). Does the utility of information influence sampling behavior? In Miyake, N., Peebles, D., and Cooper, R., editors, *Proceedings of the 34th Annual Conference of the Cognitive Science Society*, pages 719–724, Austin, TX. Cognitive Science Society.

** Awarded the Cognitive Science Society Computational Modeling Prize for higher-level cognition*

Markant, D. and Gureckis, T.M. (2012). One piece at a time: Learning complex rules through self-directed sampling. In Miyake, N., Peebles, D., and Cooper, R., editors, *Proceedings of the 34th Annual Conference of the Cognitive Science Society*, pages 725–730, Austin, TX. Cognitive Science Society.

Markant, D. and Gureckis, T.M. (2010) Category learning through active sampling. In Ohlsson, S. and Catrambone, R., editors, *Proceedings of the 32th Annual Conference of the Cognitive Science Society*, pages 248–253, Austin, TX. Cognitive Science Society.

Gureckis, T.M. and **Markant, D.** (2009) Active learning strategies in a spatial concept learning game. In Taatgen, N., van Rijn, H., Schomaker, L. and Nerbonne, J. (Eds), *Proceedings of the 31st Annual Conference of the Cognitive Science Society* (pp. 3145-3150). Austin, TX: Cognitive Science Society.

BOOK CHAPTERS

Martin, F. & **Markant, D.** (2020). Adaptive learning modules. In M. David & M. Amey (Eds.), *The SAGE encyclopedia of higher education* (Vol. 1, pp. 79-80). Thousand Oaks, CA: SAGE Publications, Inc. doi: 10.4135/9781529714395.n31

Markant, D. & Hertwig, R. (2019). Rivals in the dark: Trading off environmental and strategic

uncertainty. In Pleskac, T., Pachur, T. & Hertwig, R. (Eds), *Taming Uncertainty*. MIT Press.

Wulff, D., **Markant, D.** & Pleskac, T. (2019). Adaptive exploration in decisions from experience. In Pleskac, T., Pachur, T. & Hertwig, R. (Eds), *Taming Uncertainty*. MIT Press.

MANUSCRIPTS UNDER REVIEW OR INVITED FOR RESUBMISSION

Underlined names denote student co-authors.

Markant, D., Sah, S., Karduni, A., Dou, W. (under review). Am I right about *them*? Interactions between belief elicitation and uncertainty visualization when evaluating data about political outgroups. Submitted to VIS2026.

Glass, S., Hall, R., and **Markant, D.** (under review). Breaking Away from Destructive Feedback: The Role of Emotion Regulation in Negative Self-Verification.

Glass, S., Levens, S., and **Markant, D.** (under review). Adaptive emotion regulation and the preference for self-verifying feedback.

PREPRINTS

Markant, D., Phillips, N. D., Kareev, Y., Avrahami, J., and Hertwig, R. To act fast or to bide time? Adaptive exploration under competitive pressure.
[\[preprint\]](#)

DISSERTATION

Markant, D. (2014). *Uncertainty sampling: Computational analyses of self-directed exploration and learning* (Doctoral dissertation, New York University). Retrieved from ProQuest Dissertations and Theses.

ORAL PRESENTATIONS AT PROFESSIONAL MEETINGS

Underlined names denote student co-authors.

Markant, D. (2026). Exploring to build knowledge: Developmental Shifts in Active Learning of Transitive Relations. Talk presented at 2026 North Carolina Cognition Conference. Durham, NC.

Karduni, A., Markant, D., Sah, S., Rogha, M., Thai, M., and Dou, W. (2025). Correcting misperceptions at a glance: Using data visualizations to reduce political sectarianism. Talk presented at IEEE VIS 2025.

Markant, D., Glass, S. (2024, Nov). Emotion Regulation and the Preference for Self-verifying Feedback. Talk presented at the Annual Meeting of the Psychonomic Society. New York, NY.

Markant, D., Rogha, M., Karduni, A., Wesslen, R., and Dou, W. (2023, April). When do data visualizations persuade? The impact of prior attitudes on learning from visualizations of correlations. Talk presented the 2023 CHI Conference on Human Factors in Computing Systems (CHI 2023).

Hamburg, Germany.

Markant, D. (2022, July). Modeling the effect of chained study in transitive inference. Talk presented at the 44th Annual Meeting of the Cognitive Science Society. Toronto, Canada.

Markant, D. (2022, July). Learning the order of things: Modeling encoding- and retrieval-based strategies for transitive inference. Talk presented at the Virtual Annual Meeting of the Society for Mathematical Psychology.

Markant, D. and Ruggeri, A. (2021, November). Active control of study enhances relational learning among adolescents. Talk presented at the Annual Meeting of the Psychonomic Society (virtual).

Wesslen, R., Karduni, A., **Markant, D.**, and Dou, W. (2021, October). Effects of uncertainty visualizations on myopic loss aversion and equity premium puzzle in retirement investment decisions. Talk presented at IEEE VIS 2021.

Karduni, A., **Markant, D.**, Wesslen, R., & Dou, W. (2020). A Bayesian cognition approach for belief updating of correlation judgment through uncertainty visualizations. Talk presented at IEEE VIS 2020.

Markant, D. (2020, July). Effects of “chained” study on spontaneous relational discovery. Talk presented at the 42st Annual Meeting of the Cognitive Science Society (virtual).

Markant, D. (2020, March 21). “Chained study and spontaneous relational discovery.” Annual Meeting of North Carolina Cognition Group, Durham, NC, US. (Conference canceled)

Fantasia, V., Ruggeri, A., & **Markant, D.** (December, 2019). “Memory enhancements from active control of learning in children with Autism Spectrum Disorder.” Talk presented at the 2019 Annual Meeting of the Italian Association of Cognitive Science (AISC 2019), Rome, Italy.

Wesslen, R., Dou, W., **Markant D.**, & Shaikh, S. (October, 2019). “Human-in-the-loop: Using AI to augment human decision making, not replace it.” Bank of America 2019, Data Science & AI Conference, Charlotte, NC.

Markant, D. & Padro, M. (November, 2019). “Using metacognitive support to facilitate active learning in ADHD.” Talk presented at the Annual Meeting of the Psychonomic Society, Montreal, Canada.

Markant, D. (August, 2019). “Navigating the chain of command: Enhanced integrative encoding through active control of study.” Talk presented at the 41st Annual Meeting of the Cognitive Science Society, Montreal, Canada.

Markant, D. (March, 2019). “The emergence of an active learning advantage for episodic memory.” Talk presented at the Society for Research in Child Development (SRCD) Annual Meeting, Baltimore, MA.

Markant, D. (May, 2018). “Enhanced memory as a common effect of active learning.” Talk presented at the 30th Annual Convention of the Association for Psychological Science, San Francisco, CA.

Markant, D. (August, 2016). “Workshop: Active learning: Cognitive development, education, and computational models.” Talk presented at the 38th Annual Meeting of the Cognitive Science Society,

Philadelphia, PA.

Markant, D. (May, 2016). "Hypothesis generation sets the stage for self-directed category learning." Talk presented at the International Meeting of the Psychonomic Society, Granada, Spain.

Gureckis, T. M., Chan, P., **Markant, D.**, and Lake, B. (November 2015). "Adaptive teaching: Improving the efficiency of learning through hypothesis-dependent selection of training data." Talk presented at Annual Meeting of the Psychonomic Society, Chicago, IL, USA.

Markant, D., Pleskac, T. J., Diederich, A., Pachur, T., and Hertwig, R. (August, 2015). "Modeling choice and search in decisions from experience: A sequential sampling approach." Talk presented at the 25th Subjective Probability, Utility, and Decision Making Conference, Budapest, Hungary.

Markant, D., Phillips, N., Kareev, Y., Avrahami, Y., & Hertwig, R. (August, 2015). "Acting fast vs. biding time: Effects of competition on information sampling." Talk presented at the 25th Subjective Probability, Utility, and Decision Making Conference, Budapest, Hungary.

Markant, D., Pleskac, T. J., Diederich, A., Pachur, T., and Hertwig, R. (July, 2015). "Modeling choice and search in decisions from experience: A sequential sampling approach." Talk presented at the Annual Meeting of the Society for Mathematical Psychology, Newport Beach, CA.

Markant, D., Pleskac, T. J., Diederich, A., Pachur, T., and Hertwig, R. (July, 2015). "Modeling choice and search in decisions from experience: A sequential sampling approach." Talk presented at the 37th Annual Meeting of the Cognitive Science Society, Pasadena, CA.

de Leeuw, J.R., Coenen, A., **Markant, D.**, Martin, J.B., McDonnell, J.V., Rich, A.S., Gureckis, T.M. (July, 2014). "Workshop: Online Experiments using jsPsych, psiTurk, and Amazon Mechanical Turk" in P. Bello, M. Guarini, M. McShane, & B. Scassellati (Eds.) *Proceedings of the 36th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society.

Markant, D., Gureckis, T.M., Meder, Nelson, J.D., Pirolli, P., and Yu, C. (August, 2013) "Symposium: Informavores: Active information foraging and human cognition" in M. Knauff, M. Pauen, N. Sebanz, & I. Wachsmuth (Eds.) *Proceedings of the 35th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society.

Coenen, A., **Markant, D.**, Martin, J.B., McDonnell, J.V. (August, 2013) "Workshop: Using Mechanical Turk and PsiTurk for Dynamic Web Experiments" in M. Knauff, M. Pauen, N. Sebanz, & I. Wachsmuth (Eds.) *Proceedings of the 35th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society.

Markant, D. and Gureckis, T. M. (August, 2012). "One piece at a time: Learning complex rules through self-directed sampling." Talk presented at 34th Annual Meeting of the Cognitive Science Society, Sapporo, Japan.

Markant, D. and Gureckis, T. M. (August, 2012). "Does the utility of information influence sampling behavior?" Talk presented at the 34th Annual Meeting of the Cognitive Science Society, Sapporo, Japan.

Markant, D. and Gureckis, T. M. (August, 2010). "Category learning through active sampling." Talk

presented at the 32nd Annual Meeting of the Cognitive Science Society, Portland, OR.

Markant, D. and Gureckis, T.M. (August, 2009) "Modes of information search in active learning." Talk presented at MathPsych 2009, Amsterdam, Netherlands.

POSTER PRESENTATIONS AT PROFESSIONAL MEETINGS

Underlined names denote student co-authors.

Gordon, J. and **Markant, D.** (February, 2026). "Distrust, Deliberation, and Distress: A Dual-Process Account of Cynicism, Skepticism, and Stress." Poster to be presented at the Judgment and Decision Making Preconference, Society for Personality and Social Psychology Annual Meeting (SPSP).

Markant, D. (July, 2025). Self-verification and the perceived reliability of uncertain feedback sources. Poster presented at the 47th Annual Conference of the Cognitive Science Society. San Francisco, CA.

Hall, R., Glass, S., and **Markant, D.** (May, 2025). "Words of Choice: Exploring Emotional Regulation and the Preference for Self-verifying Feedback." Poster presented at the 2025 Meeting of the Association of Psychological Science. Washington, DC.

Hall, R., Glass, S., and **Markant, D.** (May, 2025). "Words of Choice: Exploring Emotional Regulation and the Preference for Self-verifying Feedback." Poster presented at 2025 Graduate Research Symposium, UNC Charlotte.

** Winner of 2nd place in poster competition.*

Andresen, M., and **Markant, D.** (May, 2025). "The relationship between anecdotal claims and conspiracist beliefs in evaluating online claims." Poster presented at 2025 Graduate Research Symposium, UNC Charlotte.

Gordon, J., and **Markant, D.** (May, 2025). "On cynicism: Exploring the relationship between cynicism, skepticism, and health outcomes." Poster presented at 2025 Graduate Research Symposium, UNC Charlotte.

Tsai, I., Yilmarenze, A., and **Markant, D.** (April, 2025). "How perceptual variability impacts the communication of probabilities with icon arrays." Poster presented at 2025 North Carolina Cognition Conference, UNC Chapel Hill.

Tsai, I., Yilmarenze, A., and **Markant, D.** (April, 2025). "How perceptual variability impacts the communication of probabilities with icon arrays." Poster presented at 2025 Undergraduate Research Conference, UNC Charlotte.

Glass, S., and **Markant, D.** (August, 2024) "Proposed Methodology for the Development of a Game-Based Cognitive Flexibility Training Application". Poster Presentation presented at IEEE 12th International Conference on Serious Games and Applications for Health, Funchal, Portugal.

Glass, S. and **Markant, D.** (March, 2023). "The relationship between emotion regulation and self-verification." Poster presented at 2023 Graduate Research Symposium, UNC Charlotte.

** Winner of 1st place in poster competition.*

Glass, S. and **Markant, D.** (February, 2023). "The relationship between emotion regulation and self-verification." Poster presented at 2023 North Carolina Cognition Conference, Wake Forest University.

Glass, S., Levens, S., and **Markant, D.** (February, 2023). "The relationship between emotion regulation and self-verification." Poster presented at the Annual Meeting of the Society for Affective Science: *Positive Emotion Pre-conference Workshop*. Long Beach, CA, USA.

Glass, S., Levens, S., and **Markant, D.** (March, 2022). "The relationship between emotion regulation and preferences for self-verifying feedback." Poster presented at the Annual Meeting of the Society for Affective Science (virtual).

Markant, D. (July, 2021). "Capturing uncertainty in relational learning: A Bayesian model of discrimination-based transitive inference." Poster presented at the 43rd Annual Meeting of the Cognitive Science Society (virtual).

Padro, M., Mostafavi, M. and **Markant, D.** (July, 2020). "Risk preferences and option generation: Do risk-takers generate more risky courses of action?" Poster presented at the 42st Annual Meeting of the Cognitive Science Society (virtual).

Padro, M. and **Markant, D.** (March, 2019). "Navigating the 'chain of command': How working memory impacts active learning. Poster presented at 2019 North Carolina Cognition Conference.

** Winner of 1st place in poster competition.*

Markant, D. (November, 2018). Enhanced integrative encoding through active control of learning. Poster presented at the Annual Meeting of the Psychonomic Society, New Orleans, LA.

Markant, D., Jones, A., Pachur, T., Gopnik, A., & Ruggeri, A. (July, 2018). Identifying the structure of hypotheses that guide search during development. Poster presented at the 40th Annual Meeting of the Cognitive Science Society, Madison, WI.

Kachergis, G., Adams, K., **Markant, D.** (August, 2017). "Children's familiarity preference in self-directed study improves recognition memory." Poster presented at the 39th Annual Meeting of the Cognitive Science Society. London, England, UK.

Markant, D., Pleskac, T., Diederich, A., Pachur, T., & Hertwig, R. (November, 2016). "Modeling adaptive exploration in decisions from experience: A sequential sampling approach." Poster presented at the Annual Meeting of the Psychonomic Society, Boston, MA.

Markant, D., Phillips, N., Kareev, Y., Avrahami, Y., & Hertwig, R. (November, 2015). "Acting fast vs. biding time: Effects of competition on information sampling." Poster presented at the Annual Meeting of the Society for Judgment and Decision Making, Chicago, IL, USA.

Markant, D., Pleskac, T. J., Diederich, A., Pachur, T., and Hertwig, R. (May, 2015). "Modeling choice and search in decisions from experience: A sequential sampling approach." Poster presented at the Workshop on Memory Processes in Judgment and Decision Making, Holstein, Switzerland.

Markant, D., Chan, P., Coenen, A., Martin, J., McDonnell, J., Rich, A. and Gureckis, T. M. (November

2014). "psiTurk: An open-source framework for conducting reproducible behavioral experiments online." Poster presented at the 2014 meeting of the Society for Computers in Psychology, Long Beach, CA, USA.

Chan, P., **Markant, D.**, Lake, B. and Gureckis, T.M. (July 2014). "Adaptive teaching: Improving the efficiency of learning through hypothesis-dependent selection of training data." Poster presented at the 36th Annual Meeting of the Cognitive Science Society, Quebec City, Canada.

Markant, D. and Gureckis, T.M. (August 2013). "Adapting search based on hypothesis space sparsity." Poster presented at the 35th Annual Meeting of the Cognitive Science Society, Berlin, Germany.

Markant, D., Dubrow, S., Davachi, L., Gureckis, T.M. (August 2012). "The role of exploratory decision-making in enhancing episodic memory." Poster presented at the 34th Annual Meeting of the Cognitive Science Society, Sapporo, Japan.

Gureckis, T. M. and **Markant, D.** (November, 2011). "How search in the mind guides search in the world." Poster presented at the Annual Meeting of the Psychonomic Society, Seattle, WA.

Markant, D. and Gureckis, T. M. (August, 2011). "Sampling at the Margins to Learn Perceptual Categories." Poster presented at the 33rd Annual Meeting of the Cognitive Science Society, Boston, MA.

Markant, D. and Gureckis, T. M. (November, 2010). "An Exploration of Active Sampling in Perceptual Classification Tasks." Poster presented at Annual Meeting of the Psychonomic Society, St. Louis, MO.

Markant, D. and Gureckis, T. M. (August, 2010). "Discovering complex categories through active sampling." Poster presented at IEEE International Conference on Development and Learning, Ann Arbor, MI.

Gureckis, T.M. and **Markant, D.** (2009) "Active Learning Strategies in a Spatial Concept Learning Game." Poster presented at the 31st Annual Conference of the Cognitive Science Society, Amsterdam, Holland.

INVITED PRESENTATIONS

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| April, 2023 | Markant, D. (2023, April 4). "Learning the order of things: How people construct relational knowledge through active exploration." Psychology Research Seminar, Davidson College. |
| October, 2022 | Markant, D. (2022, October 16). "When does the data speak for itself? Toward formal models of persuasive data visualization." Keynote presentation at TREX 2022: Workshop on Trust and Expertise in Visualization, IEEE VIS 2022. |
| September, 2022 | Markant, D. (2022, September 8). "When does the data speak for itself? Effects of prior attitudes and beliefs on learning from data visualizations." Cognitive Seminar, UNC Chapel Hill. |
| August, 2020 | Markant, D. (2020, August). "Constructing relational knowledge from experience: Active control in transitive inference." The Reasoning Lab, Naval Research Laboratory. |

- December, 2019 "A new curriculum on open and transparent practices in Research Methods," with Dr. Alexia Galati. 2019 Winter Meeting of the Charlotte Area Network for the Teaching of Psychology (CANTOP), Charlotte, NC.
- October, 2019 "Human-in-the-Loop: Using AI to augment human decision making, not replace it." Panel discussion at 2019 Data Science and AI Conference, Bank of America, Charlotte, NC.
- October 2018 "Enhancing memory through active control." University of North Carolina at Greensboro.
- November, 2016 "Uncertainty-driven exploration." *Summit on Human Problem Solving and Artificial Intelligence*, Sandjerg, Denmark.
- October, 2016 "Constructing rules through active exploration: Evidence from human category learning." *Neurocuriosity 2016: Second Interdisciplinary Symposium on Information-seeking, Curiosity and Attention*, London, UK.
- August, 2016 "CHASE-ing the dream: A sequential sampling framework for modeling decisions from experience." *4th Summer School on Decisions from Experience*, Warwick Business School.
- April, 2016 "On choosing your own adventure: Effects of active control on learning and memory." Affective Brain Lab, University College London.
- November, 2015 "Hypothesis generation sets the stage for self-directed learning." Experimental Psychology Seminar, University College London.
- March, 2015 "Acting fast vs. biding time: Exploration in competitive environments." Center for the Study of Rationality, Hebrew University of Jerusalem.
- December, 2013 "Bayesian models of self-directed information gathering" (with Todd Gureckis). BBN Workshop on Bayesian Cognitive Modeling for the Intelligence Community, BBN Raytheon.
- July, 2013 "What you don't know can help you: Uncertainty-driven sampling during learning." Center for Adaptive Rationality, Max Planck Institute for Human Development.
- May, 2013 "Into the unknown: How uncertainty guides active learning." University Seminar Series, Stevens Institute of Technology.
- April, 2013 "Decomposing the 'active' learner: Effects of self-directed control on learning and memory." Department of Psychological and Brain Sciences, Indiana University.

AWARDS AND HONORS

2019	Catalyst Program Fellow, University of North Carolina at Charlotte
2014	Psychonomic Society Clifford T. Morgan Best Article Award (<i>Memory & Cognition</i>)
2013	American Psychological Association Dissertation Research Award
2013	Learning Analytics Summer Institute (LASI), Stanford University
2012	Computational Modeling Prize in Higher-level Cognition for paper presented at the 2012 Meeting of the Cognitive Science Society
2012	Robert J. Glushko and Pamela Samuelson Foundation Student Travel Grant for attendance to the 2012 Meeting of the Cognitive Science Society
2011	Graduate Summer School: “Probabilistic Models of Cognition”, Institute of Pure and Applied Mathematics, UCLA
2010	Graduate School of Arts and Sciences Travel Award, NYU
2010	Honorable Mention, NSF Graduate Research Fellowship Program
2008—2013	MacCracken Fellowship, NYU Graduate School of Arts and Sciences

EXTERNAL GRANT PROPOSALS

Not funded

NSF: Research on Innovative Technologies for Enhanced Learning (RITEL) Program
Title: “Foundation Models for Socratic Teaching of Humans and Machines”
Co-investigator; \$617,102 direct costs requested
Submitted: January 2024

DOD Department of Army
Title: “Toward Seamless Human-Automation Handovers -- An Adaptive Haptic Shared Steering Control and Path Planning Paradigm for Ground Vehicles”
Co-investigator; \$373,786 direct costs requested
Submitted: November 2022

NSF: Research Traineeship (NRT) Program
Title: “Trustworthy Human-AI Interdisciplinary Consortium (THInC)”
Co-investigator; \$2,263,018 direct costs requested
Submitted: September 2022

NSF: AI Research Institutes
Title: “Advancing the Neurocognitive Foundations of AI through Human Centered Creative AI”
Co-investigator; \$17,787,403 direct costs requested
Submitted: May 2022

NIH: National Institute of Child Health and Human Development (NICHD)

Title: "Engaging endogenous selective attention control to promote children's learning from relevant contextual information"

Co-investigator; \$1,827,875 direct costs requested

Submitted: October 2021

NSF: Research on Emerging Technologies for Teaching and Learning (RETTL)

Title: "Privacy-preserving Infrastructure for Personalized Dialogue System to Support Self-Regulated Learning in Computer Science"

Co-investigator; \$590,362 direct costs requested

Submitted: October 2021

IARPA: Better Extraction from Text Towards Enhanced Retrieval (BETTER)

Title: "PRIMD: Personalized Retrieval of Information in Multilingual Data"

Co-investigator

Submitted: September 2018

INTERNAL GRANT PROPOSALS

Funded

UNC Charlotte Data Science Initiative Seed Grant

Title: "Curating Heterogenous, Unstructured Health Data to build Personalized Health Infrastructures and Digital Assistants."

Co-investigator; \$30,000 requested

Submitted: December 2019

Award amount: \$10,000 (2020)

UNC Charlotte Scholarship of Teaching and Learning (SoTL) grant

Title: "Examining the impact of open and transparent research practices in an undergraduate research methods course in Psychology"

Co-PI; \$15,000 requested

Submitted: November 2018

Award amount: \$15,000 (2019–2020)

UNC Charlotte Faculty Research Grant

Title: "Executive functioning and active learning: The effects of impaired cognitive function on self-directed inquiry."

PI; \$8,000 requested

Submitted: October 2018

Award amount: \$8,000 (2019–2020)

Not funded

University of North Carolina Undergraduate Research Program

Title: "Introduction to Open Science" for undergraduates: Teaching the foundations of transparent and reproducible research

Co-investigator; \$25,238 direct costs requested

Submitted: September 2019

PROFESSIONAL SERVICE

Committees	Organizing Committee, Annual Meeting of the Cognitive Science Society (2020, 2021, 2022) Organizing Committee, Annual Meeting of the Society for Text & Discourse (2026) Program Committee, Annual Meeting of the Cognitive Science Society (2020, 2021, 2022, 2023, 2024, 2025, 2026) Program Committee, Visualization for Social Good Workshop at VIS2023 (2023)
Journal reviewing	<i>Annual Meeting of the Cognitive Science Society</i> (2009, 2010, 2013–2023) <i>Behavior Research Methods</i> (2014, 2015, 2018) <i>Child Development</i> (2021, 2022) <i>Child Development Perspectives</i> (2025) <i>Cognition</i> (2015, 2016, 2018, 2020, 2021, 2023, 2024, 2025) <i>Cognitive Psychology</i> (2016, 2019, 2020, 2022) <i>Cognitive Science</i> (2013, 2014, 2016, 2018, 2019, 2020, 2021, 2022, 2025) <i>Collabra: Psychology</i> (2024, 2025) <i>Nature Communications: Psychology</i> (2024) <i>Consciousness and Cognition</i> (2019) <i>Cortex</i> (2016) <i>Decision</i> (2014, 2017) <i>Developmental Psychology</i> (2016) <i>Developmental Science</i> (2025) <i>Frontiers in Psychology</i> (2020, 2021, 2022, 2023) <i>IEEE VIS (Visualization and Visual Analytics)</i> (2024, 2025) <i>Journal of Cognition and Development</i> (2016) <i>Journal of Cognitive Psychology</i> (2014) <i>Journal of Economic Psychology</i> (2020) <i>Journal of Experimental Child Psychology</i> (2024) <i>Journal of Experimental Psychology: General</i> (2015, 2016, 2017, 2019, 2021, 2022, 2023, 2024) <i>Journal of Experimental Psychology: Human Perception & Performance</i> (2015) <i>Journal of Experimental Psychology: Learning, Memory & Cognition</i> (2015) <i>Journal of Mathematical Psychology</i> (2014, 2016) <i>Judgment & Decision Making</i> (2022) <i>Learning & Memory</i> (2021) <i>Management Science</i> (2018) <i>Memory & Cognition</i> (2013, 2014, 2017, 2025) <i>Nature Human Behavior</i> (2025) <i>Nature Machine Intelligence</i> (2019) <i>Open Mind</i> (2018, 2019, 2020) <i>PLOS Computational Biology</i> (2014, 2022) <i>PLOS One</i> (2016) <i>Psychological Science</i> (2015, 2017, 2018, 2019)

Psychological Research (2022)
Psychological Review (2022, 2024)
Psychonomic Bulletin and Review (2010, 2019, 2020, 2021, 2022, 2024)
Psychophysiology (2023)
Quarterly Journal of Experimental Psychology (2022)
Thinking & Reasoning (2021)
Topics in Cognitive Science (2017, 2018)
Trends in Cognitive Science (2014)

Grant reviewing *National Science Foundation: Graduate Research Fellowship Program* (2022)
APA/CDC: "Psychological science addressing COVID-19 pandemic priorities" (2021)
Israel Science Foundation (2021)
U.S. Department of Defense (2018)
Estonian Research Council (2017)
National Science Foundation: Decision, Risk, and Management Science (2015, 2016)
National Science Foundation: Cognitive Neuroscience (CAREER) (2024)

Editorial boards *Frontiers in Psychology (Cognition: Review editor, 2020—present)*
Peer Community in Registered Reports (2023—present)
Peer Community in Psychology (2025—present)

Memberships Cognitive Science Society
Psychonomic Society
Society for Mathematical Psychology
SPARK Society
Association for Computing Machinery (ACM)

Other service Psychonomic Society Graduate Conference Award Review Committee (2021)
Mentor, CogSci mentoring program (2020, 2021)

SERVICE TO DEPARTMENT, COLLEGE, and UNIVERSITY

Department of Psychological Science and Affiliated Graduate Programs

Graduate Program Director, Graduate Certificate in Cognitive Science (Fall 2020—present)
Associate Program Director, Health Psychology Ph.D. program (Fall 2024—present)
Member, Dept. of Psychological Science I/O Tenure-track Faculty Search Committee (2024–2025)
Member, Dept. of Psychological Science Advisory Committee (2024–2026)
Chair, Health Psychology Experimental Research Seminar planning committee (2022–2023)
Member, Health Psychology Diversity, Equity, and Inclusion Committee (2022–2024)
Member, Health Psychology Experimental Research Seminar planning committee (2021–2022)
Member, Health Psychology Research Series planning committee (2020–2022)
Member, Health Psychology Showcase planning committee (2022–2023)
Member, Health Psychology Climate Survey Recommendation committee (2021–2022)
Member, Psychology M.A. program Advisory Committee (2020–2021)
Member, Lecturer Reappointment Guidelines Committee (2019)

College and University Service

Faculty judge for Graduate Research Symposium (2022, 2023, 2024, 2025)

Member, Psychological Science Department Chair Search Committee (2022)

Member, CLAS Racial and Social Justice Committee (Graduate Student Success) (2020-2021)

GRADUATE THESIS AND DISSERTATION COMMITTEES

Chair

Ryan Hall (Ph.D. Health Psychology, ongoing)

Josh Gordon (Ph.D. Health Psychology, ongoing)

Noah Mangione (M.A. Psychology, ongoing)

Isabella Tsai (M.A. Psychology, ongoing)

Ata Yilmazemre (M.A. Psychology, ongoing)

Shaina Glass: *"Fun with Cognitive Flexibility: The Development and Evaluation of a Game-based Application"* (Ph.D. Health Psychology; completed 2025)

Matt Andresen: *"The relationship between anecdotal claims and conspiracist beliefs in evaluating online claims"* (M.A. Psychology; completed 2025)

Meagan Padro: *"Using metacognitive guidance to facilitate active learning in ADHD"* (M.A. Psychology; completed 2019).

Member

Subham Sah (Ph.D. Computer Science, ongoing)

Milad Rogha (Ph.D. Computer Science, completed 2026)

Daniel Perkins (M.A. Psychology, completed 2026)

Jeba Rezwana (Ph.D. Computer Science, completed 2023)

Philip Zendels (Ph. D. Health Psychology, completed 2023)

Ali Algarni (Ph.D. Computer Science, completed 2022)

Safat Siddiqui (Ph.D. Computer Science, completed 2022)

Ryan Hall (M.A. Psychology, completed 2022)

Jingoo Kim (Ph.D. Computer Science, completed 2022)

Ryan Wesslen (Ph. D. Computer Science, completed 2021)

Michael Brunswick (M.A. Psychology, completed 2021)

Tevin Williams (M.A. Psychology, completed 2021)

Alireza Karduni (Ph. D. Computer Science, completed 2021)

Matthew Granson (M.A. Psychology, completed 2020)

Lauren Johnson (M.A. Electrical Engineering, completed 2019)

UNDERGRADUATE HONORS THESES

Chair

Greyson Shafiei: *"Emotion recognition through GPT-4 computer vision analysis of facial expressions."* (Psychology and Computer Science, completed 2025)

Cooper Davis: *"Flow, performance, and memory: An examination of the relationship between flow state and episodic memory during computer game play."* (Psychology, completed 2022)

Arielle Little: *"Do active learning strategies benefit students with ADHD?"* (Psychology, completed 2019)

Member

Alex Litovchenko (Psychology, completed 2025)

Ata Yilmarenze (Psychology, completed 2025)

Evelyn Valenzuela (Psychology, completed 2025)
 Justin Castillo (University Honors, completed 2022)
 Ricky Rodriguez-Cue (Psychology, completed 2022)
 Philip Hinson (Psychology, completed 2019)

TEACHING AND INSTRUCTIONAL ACTIVITIES

Courses taught at UNC Charlotte

Undergraduate	PSYC2103 — Research Methodology II Fall 2017, Spring 2018, Fall 2018, Fall 2019, Spring 2020, Spring 2021
	PSYC3111 — Psychology of Learning and Memory Spring 2019, Spring 2020, Fall 2020, Spring 2021, Fall 2021, Spring 2022, Spring 2024, Fall 2024, Fall 2025
Graduate	PSYC/ITIS/ITCS 6216 — Introduction to Cognitive Science Spring 2018, Fall 2018, Fall 2019, Fall 2020, Fall 2021, Fall 2022, Fall 2024, Fall 2025
	PSYC/ITCS 6617 — Computational Human Behavior Modeling Spring 2023, Spring 2025
	HPSY 8601 — Foundations in Psychology I: Physiological, Cognitive, and Affective Bases of Behavior Fall 2022, Fall 2024, Spring 2025
	PSYC 6261 — Independent Study in Health Psychology (Experimental Research Seminar) Fall 2022

Guest lectures, seminars and workshops presented at UNC Charlotte

September 2024	Markant, D. (2024, September 11). "Cognitive Theories of Communication." Workshop for graduate students enrolled in Cognitive Science of Communication course.
February 2023	Markant, D. (2023, February 17). "Cultivating a professional online presence." Health Psychology Experimental Research Seminar, UNC Charlotte.
April 2022	Markant, D. (2022, April 29). "How interactions with data drive belief updating and persuasion." College of Computing and Informatics Graduate Seminar, UNC Charlotte.
April 2022	Markant, D. and Galati, A. (2022, April 8). "Towards open and transparent research practices." Health Psychology Research Series, UNC Charlotte.

- March 2022 **Markant, D.** and Galati, A. (2023, March 25). "Assessing the impact of curriculum on open and transparent research practices." *Scholarship of Teaching and Learning Research Showcase* (video series), Center for Teaching and Learning, UNC Charlotte.
- March 2022 **Markant, D.** (2022, March 21). "Towards formal models of persuasion during interactive data visualization." Ribarsky Center for Visual Analytics, UNC Charlotte.
- March 2020 **Markant, D.** and Galati, A. (2023, March 25). "Assessing the impact of a new curriculum on open and transparent research practices." Research Showcase, Office of Assessment and Accreditation, UNC Charlotte.
- May 2019 **Markant, D.** (2019, May 18). "The probabilistic mind: Introduction to Bayesian cognitive modeling." Guest lecture for *PSYC/ITSC 6617 Computational Human Behavior Modeling*.
- May 2018 **Markant, D.** (2018, May 2). "Using PollEverywhere to Engage and Reactivate Knowledge." *Active Learning Academy Spring 2018 Expo*, Center for Teaching and Learning, UNC Charlotte.